# A Bit Inside

*See what's happening inside the technology you use every day.*

**DAILY CONTENT PLAN — SUNDAY, OCTOBER 11, 2026**

5 short-form videos · YouTube Shorts · TikTok · Facebook Reels

Slots (Yangon): 8:00 AM · 11:30 AM · 2:30 PM · 5:30 PM · 8:30 PM

# DAILY CONTENT PLAN — SUNDAY, OCTOBER 11, 2026

**A Bit Inside** · Technology & Digital Life · *"See what's happening inside the technology you use every day."* Platforms: YouTube Shorts · TikTok · Facebook Reels · all videos 1080×1920, 30 fps, H.264 + AAC, −14 LUFS, 60–180 s, each with a custom 9:16 cover (PNG for TikTok/Facebook, JPG under 2 MB for YouTube).

**Status:** all 5 videos produced and checked (READY-01 … READY-05). Nothing has been uploaded, scheduled or posted.

**Mix today:** 2 evergreen + 3 emerging, leaning emerging as planned yesterday to bring the 3-day average back toward 70/30:

- **Emerging:** 01 is this week's news (Honda wireless-charging road, Oct 5, 2026). 03 is built on fresh data (Imperva's 2026 Bad Bot Report, Apr 29, 2026). 05 is news-led (Voyager 1 reaches one light-day on Nov 18, 2026, per NASA).
- **Evergreen:** 02 (diffusion) and 04 (Year 2038). 04 carries a real, fixed deadline and 2020–2025 fixes, so it still feels current.

The Tavus "Griffin" candidate was dropped: it's a small company-run test (54 testers), too thin to build a verified video on.

| Slot (Yangon) | # | Topic | Category · Format | Type | Runtime | Hook (0–2 s) | Overall viral |
|---|---|---|---|---|---|---|---|
| 8:00 AM | 01 | This road charges your EV while you drive (Honda + Taisei) | Tech news fact · side-view road + cross-section | Emerging (Honda, Oct 5, 2026) | 1:11.7 | "Honda is testing a road that charges electric vehicles while they drive." | 86 |
| 11:30 AM | 02 | This AI image started as pure static (diffusion) | AI explainer · problem → explanation → takeaway | Evergreen | 1:27.0 | "This AI image started as pure TV static." | 85 |
| 2:30 PM | 03 | Most web traffic isn't human (bots 53%, humans 47%) | Curiosity "wait… WHAT?" · Progressive Reveal | Emerging (Imperva 2026 report, Apr 29, 2026) | 1:25.3 | "Most of the traffic on the web isn't human." | 87 |
| 5:30 PM | 04 | In 2038, some computers will think it's 1901 (Year 2038 problem) | **Flagship** · systems story: count → overflow → real crash → risk → fix | Evergreen (fixed 2038 deadline, 2020–2025 fixes) | 2:26.6 | "In 2038, some computers will think it's 1901." | **89** |
| 8:30 PM | 05 | Voyager 1 will soon be one light-day from home | Deep / reflective · starfield, calm closed ending | Emerging (NASA milestone, Nov 18, 2026) | 1:33.3 | "On November 18th, something we built will be one full light-day from home." | 86 |

## Scores (1–100)

| # | Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|---|
| 01 Road charges EV | 88 | 87 | 82 | 85 | 84 | 84 | 92 | **86** |
| 02 AI images from noise | 89 | 88 | 83 | 84 | 81 | 88 | 72 | **85** |
| 03 Bots outnumber humans | 91 | 90 | 84 | 88 | 87 | 84 | 84 | **87** |
| 04 Year 2038 problem | 93 | 92 | 87 | 89 | 86 | 89 | 80 | **89** |
| 05 Voyager one light-day | 87 | 85 | 84 | 88 | 82 | 85 | 89 | **86** |

## Files

| # | Script | Captions | Video | Thumbnail |
|---|---|---|---|---|
| 01 | `01-road-charges-ev.md` | `01-road-charges-ev-captions.md` | `video/01-road-charges-ev-final.mp4` | `thumbnails/01-road-charges-ev-thumb.png` / `.jpg` |
| 02 | `02-ai-images-from-noise.md` | `02-ai-images-from-noise-captions.md` | `video/02-ai-images-from-noise-final.mp4` | `thumbnails/02-ai-images-from-noise-thumb.png` / `.jpg` |
| 03 | `03-bots-outnumber-humans.md` | `03-bots-outnumber-humans-captions.md` | `video/03-bots-outnumber-humans-final.mp4` | `thumbnails/03-bots-outnumber-humans-thumb.png` / `.jpg` |
| 04 | `04-year-2038-problem.md` | `04-year-2038-problem-captions.md` | `video/04-year-2038-problem-final.mp4` | `thumbnails/04-year-2038-problem-thumb.png` / `.jpg` |
| 05 | `05-voyager-one-light-day.md` | `05-voyager-one-light-day-captions.md` | `video/05-voyager-one-light-day-final.mp4` | `thumbnails/05-voyager-one-light-day-thumb.png` / `.jpg` |

## Final checks

- **Runtime:** every video is between 60 and 180 s (ffprobe): 71.7 / 87.0 / 85.3 / 146.6 / 93.3 s. Each also fits its slot brief (03: 80–110, 04: 110–150, 05: 80–120).
- **Loudness:** −14.1 / −13.9 / −14.1 / −13.8 / −14.0 LUFS integrated.
- **Layout:** the self-check passed on all 5 videos and all 5 thumbnails ( `video/layout-check.md` ), with 0 failures:
- Titles, captions, CTA and wordmark are centered at 540.0 (±2 px).
- Everything sits inside x 120–960, visuals stay above y 1420, and nothing goes below y 1610.
- No missing glyphs: icons are drawn as vector paths, and there are no emoji in the videos.
- No title ends on a single-word line.
- The exported-mp4 pixel spot checks are within margins. The three off-center readings (01 at 46.0 s, 02 at 17.8 s, 04 at 7.4 s) are explained in the report as left-to-right fills or below-

threshold dark areas; the rendered-frame checks are 540.0.
- Contact sheets of the final mp4s were viewed.
- **Hooks:** 01–04 hooks land in 0–2 s. 05's calm hook sentence runs about 4 s by design for the reflective slot, but its visual (light pulse, Voyager and Earth) is on screen from frame 1. There's no forced like/follow in the voiceover except 04's natural flagship call to action, and end cards are visual only.
- **Sourcing:** every claim is sourced in its script file, with accuracy notes:
- 03: Imperva / Thales 2026 Bad Bot Report and blog
- 04: Wikipedia Y2038, the LKML y2038 pull, Debian 13 release notes, and the archived AOLserver report; date math is our own
- 05: NASA's light-day, "Where are Voyager 1 and 2 now", FAQ, Voyager 1, Pale Blue Dot and Golden Record pages
- All drawings are our own illustrative mock-ups, labelled as such in the descriptions.
- **No repeats:**
- Oct 8: tap-to-pay, touchscreens, DNS, iPhone aperture, GPT-6 UI
- Oct 9: QR damage, DNS, GPS + Einstein, AI agents, digital afterlife
- Oct 10: Google Playground, AI tokens, file deletion, undersea cables, link rot
- The five topics today don't overlap each other.
- **Brand:** only "A Bit Inside" branding is used. There are no third-party logos (company and agency names appear as plain text only).

# Slot 01 · 8:00 AM (Yangon) · Discovery / Tech-News Fact

## This Road Charges Your EV While You Drive (Honda + Taisei)




Cover thumbnail
`thumbnails/01-road-charges-ev-thumb.jpg`

| Field | Detail |
|---|---|
| **Topic** | Honda R&D, Taisei and Taisei Rotec announced (Oct 5, 2026) the core technology for a road-embedded wireless system that charges EVs while they drive |
| **Category** | Tech news · Emerging (6 days old) |
| **Format** | One surprising fact, animated side-view road + cross-section explainer |
| **Runtime** | 1:11.7 (71.7 s) · ffprobe-verified · within the 60–180 s rule |
| **Video** | `video/01-road-charges-ev-final.mp4` (1080×1920, 30 fps, H.264 + AAC, −14 LUFS) |
| **Thumbnail** | `thumbnails/01-road-charges-ev-thumb.png` (1080×1920) · `thumbnails/01-road-charges-ev-thumb.jpg` (YouTube, 126 KB) |

## Why this topic

It's this week's most "wait, the ROAD does it?" tech story: a real automaker has built and road-tested hardware that sends power up through the pavement into a moving vehicle. It's one clear surprising fact, it's easy to animate, and it has an honest catch (not on public roads yet), so it's not hype. It doesn't overlap with Oct 8–10 topics or with today's 02 (AI images).

## Hook (0–2 s)

- **Spoken:** "Honda is testing a road that charges electric vehicles while they drive."
- **On-screen:** THIS ROAD CHARGES YOUR EV
- **Visual:** a car drives along a road with glowing buried units; each unit lights up as the car passes over, field rings pulse up into the car, and a big battery above fills (20% → 85%).

## Full voiceover script

**[0:00 Hook]** Honda is testing a road that charges electric vehicles while they drive.

**[0:04 What was announced]** On October fifth, Honda and the Japanese builders Taisei and Taisei Rotec announced the core technology for it. No plug, and no stopping.

**[0:13 How]** Here's the trick. Units buried in the lane send power up through a magnetic field, to a receiver under the vehicle. No contact at all. It's the same basic idea as a wireless phone charger, scaled up for the road.

**[0:27 The world first]** Honda says each road unit is a world first: the coil and its inverter in one box, all fed by a shared DC power line. Less wiring, and units that fit into strips cut into existing roads.

**[0:41 Built tough]** Next, a test track will run heavy vehicles over them, the equivalent of one million wheel passes. The goal: up to one hundred fifty kilowatts, at high speed.

**[0:52 The catch]** The catch: it's not on public roads yet. Expressway tests in Chiba, Japan, start from the 2027 fiscal year, and trucks come first.

**[1:02 Ending]** If it works, the charging station of the future might not be a place you stop. It might be the road itself.

## Visual direction

| Time | Visual |
|---|---|
| 0:00–0:04 | Car drives across a road with five buried units; the unit under it glows orange, field rings pulse up, battery above fills with a % counter. |
| 0:04–0:13 | "OCT 5, 2026" pill, "HONDA · TAISEI · TAISEI ROTEC"; NO PLUG tile (plug + red × drawn as vector) appears on the axis, then slides left as the NO STOPS tile (charging station + ×) arrives. |
| 0:13–0:27 | Cross-section: large car above the road, one buried ROAD UNIT, RECEIVER pad under the car; animated magnetic field loops and energy dots rise; "NO CONTACT" pill; then "LIKE A PHONE CHARGER, SCALED UP" panel with a phone-on-pad icon. |
| 0:27–0:41 | "ONE ROAD UNIT" box: COIL (spiral) appears on the axis then slides left as INVERTER (chip) arrives, with a "+"; "WORLD FIRST, SAYS HONDA" pill; a road strip with four units joined by a yellow "SHARED DC LINE"; "FITS EXISTING ROADS" pill. |
| 0:41–0:52 | Truck on a scrolling road; a mono counter runs 0 → 1,000,000 "WHEEL PASSES"; a power bar fills to "UP TO 150 KW". |
| 0:52–1:02 | Three pills stack: NOT ON PUBLIC ROADS YET · EXPRESSWAY TESTS: FY2027+ · TRUCKS FIRST. |
| 1:02–1:12 | A charging-station icon gets a red × and fades into a full battery; the car drives along the glowing road; visual end card (follow pill). |

## On-screen text

THIS ROAD CHARGES YOUR EV → NO PLUG. NO STOPPING. → POWER THROUGH A MAGNETIC FIELD → COIL + INVERTER IN ONE ROAD UNIT → BUILT FOR TRUCKS: UP TO 150 KW → THE CATCH: NOT ON PUBLIC ROADS YET → THE ROAD IS THE CHARGER

## Ending

A natural closing thought (the charging station might be the road itself). No like/follow ask in the voiceover; the end card is visual only.

## Thumbnail

"THIS ROAD / CHARGES CARS" (white/cyan/yellow) over a glowing battery, a car on a road of glowing buried units with field rings under it. Self-check passed (lines at 540.0, visual box 182–898).

## Captions & hashtags (all 3 platforms)

8:00 AM slot (Yangon) · Sun 11 Oct 2026 · A Bit Inside

Video: /workspace/abitinside/2026-10-11/video/01-road-charges-ev-final.mp4 (1:11.7, 1080x1920, H.264 + AAC)
Cover: /workspace/abitinside/2026-10-11/thumbnails/01-road-charges-ev-thumb.png (TikTok/Facebook) · 01-road-charges-ev-thumb.jpg (YouTube)

### TikTok (@abitinside)

This road charges your EV while you drive 🔋🛣️ Honda just showed road units that beam power up through a magnetic field. No plug, no stopping. The catch: public expressway tests only start from 2027. Would you trust a road to charge your car?

#ev #electriccars #technews #honda #learnontiktok #ABitInside

### YouTube Shorts

Title: This Road Charges Your EV While You Drive (Honda's Wireless Road)

Description:
On Oct 5, 2026, Honda R&D, Taisei and Taisei Rotec announced the core technology for a road that charges EVs while they drive. Units buried in the lane send power through a magnetic field to a receiver under the vehicle, with no contact. Honda says each unit is a world first, with the coil and inverter in one box on a shared DC line, and the units fit into strips cut into existing roads. Next: a durability test equal to 1 million wheel loads, with a goal of up to 150 kW at high speed. Expressway tests in Chiba, Japan, start from fiscal 2027, trucks first. (Car, truck and road animations are illustrative, not Honda designs.)

Sources: Honda Global press release (Oct 5, 2026) · Honda News (Oct 6, 2026) · Nippon.com / Jiji Press (Oct 5, 2026)

#Shorts #EV #TechNews #ABitInside

### Facebook Reels (A Bit Inside Page)

What if the road itself charged your electric car? Honda and two Japanese construction firms just revealed road units that send power up through a magnetic field into a moving vehicle. No plug, no stopping. Here's how it works, and the catch. Would you use it?

#ABitInside #EV #TechNews

## Sources


- Honda Global, "Honda Develops Underlaying Technology for In-motion Wireless Charging of EVs, Including Large Commercial Vehicles," Oct 5, 2026 (Honda R&D with Taisei and Taisei Rotec; magnetic coupling WPT road system; GA units embedded in road, VA receiver unit on vehicle; inverter + coil integrated in each unit, "world's first" publicly disclosed as of end of Sept 2026 per Honda's research; DC distribution simplifies wiring; milling-and-embedding installation in existing pavement; withstands ~20-ton vehicles; test roadway at T-FIELD/TAMURA from late 2026 with a loading test equivalent to 1 million wheel loads at 49 kN per wheel; later verification up to 150 kW and at high speeds; Tateyama Project expressway demos (NEXCO East, Chiba) from Japanese FY2027, starting April 1, 2027; practical use first in logistics/transport): https://global.honda/en/topics/2026/c_2026-10-05beng.html
- Honda News (US), same release, Oct 6, 2026: https://hondanews.com/en-US/releases/honda-develops-underlaying-technology-for-in-motion-wireless-charging-of-evs-including-large-commercial-vehicles
- Nippon.com / Jiji Press, "Honda, Taisei Jointly Develop Wireless EV Charging System," Oct 5, 2026 (road-embedded wireless system to charge EVs while driving; expressway tests in Chiba from FY2027 or later; aimed at logistics and transport first): https://www.nippon.com/en/news/yjj2026100500880/
- AutoNext, "Honda's wireless road wants to charge moving trucks at 150 kW" (receiver under the vehicle picks up energy by magnetic coupling with no physical contact; units fit strips milled from existing pavement): https://www.autonext.co/news/honda-wireless-charging-road-150-kw-trucks-tateyama-2027

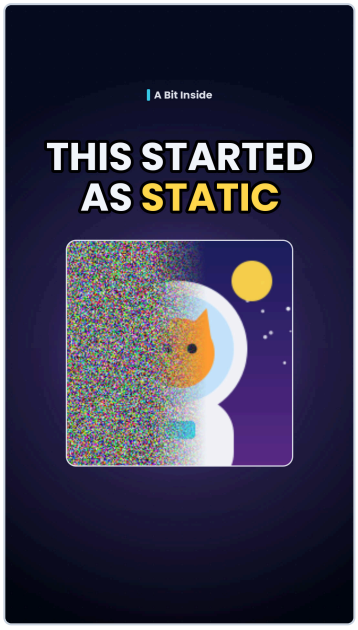

**Accuracy notes:** "World first" is Honda's own claim (first such unit publicly disclosed as of end of September 2026, per Honda's research) and is attributed to Honda in the voiceover and on screen. "One million wheel passes" simplifies Honda's "loading test equivalent to 1-million-wheel loads (49 kN per wheel)". "Up to 150 kW at high speed" is a stated future verification goal, not an achieved result, and the script says "the goal". The phone-charger comparison is our simplification: both use magnetic (inductive) coupling, but the road system is a much higher-power dynamic version. "Honda is testing" refers to the joint test-roadway validation at T-FIELD/SATTE described in the release. All car, truck, road and unit drawings are illustrative animations, not Honda designs; no logos are used. The battery percentage is illustrative.

## Scores (1–100)

| Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|
| 88 | 87 | 82 | 85 | 84 | 84 | 92 | **86** |

# Slot 02 · 11:30 AM (Yangon) · AI Explainer

## This AI Image Started as Pure Static: How Diffusion Works




Cover thumbnail
`thumbnails/02-ai-images-from-noise-thumb.jpg`

| Field | Detail |
|---|---|
| **Topic** | How AI image generators (diffusion models) turn random noise into a picture |
| **Category** | AI explainer · problem → explanation → takeaway |
| **Format** | Animated "noise → picture" demo with our own illustrative image (vector cat astronaut), step-by-step |
| **Runtime** | 1:27.0 (87.0 s) · ffprobe-verified · within the 60–180 s rule |
| **Video** | `video/02-ai-images-from-noise-final.mp4` (1080×1920, 30 fps, H.264 + AAC, −14 LUFS) |
| **Thumbnail** | `thumbnails/02-ai-images-from-noise-thumb.png` (1080×1920) · `thumbnails/02-ai-images-from-noise-thumb.jpg` (YouTube, 274 KB) |

## Why this topic

Almost everyone has seen AI images, but almost no one knows they start as random static. That's a strong "wait, what?" problem with a clean two-step explanation (add noise in training, remove it when generating) and a practical takeaway (why the same prompt gives different pictures, and what a "seed" is). It's fresh for the channel: Oct 10's 02 explained text tokens (how chatbots read), and this is about images. It doesn't overlap today's 01 (wireless EV road).

## Hook (0–2 s)

- **Spoken:** "This AI image started as pure TV static."
- **On-screen:** IT STARTED AS PURE STATIC
- **Visual:** a square of colorful static clears in about 2 seconds into a cat in a space suit, with a "NOISE 100% → 0%" meter.

## Full voiceover script

**[0:00 Hook]** This AI image started as pure TV static.

**[0:03 The problem]** So how does a computer turn random noise into, say, a cat in a space suit? The trick is called diffusion, and it's behind many popular image generators.

**[0:13 Step 1: training]** Step one happens during training. Take a real picture, and add a little random noise. Then more, and more, until nothing is left but static.

**[0:23 The model's job]** At every level of noise, the model practices one job: look at the noisy picture, and guess which noise was added. Repeat that across a huge collection of images, and it gets very good at spotting noise.

**[0:36 Step 2: generating]** Step two is making a new image. Start with fresh, random static. The model predicts the noise, removes a little, and does it again, often dozens of times. Big shapes appear first, then edges, then fine details.

**[0:52 The prompt]** Your prompt is the steering wheel. A text encoder turns your words into numbers, and at every step, those numbers nudge the cleanup toward cat, and toward space suit.

**[1:03 The seed]** That's why the same prompt gives you a different picture every time. Each run starts from different static. Reuse the same starting noise, called a seed, with the same settings, and you can usually get the same image back.

**[1:17 Takeaway]** So an AI image isn't fetched from a folder of pictures. It's carved out of noise, one tiny cleanup at a time.

## Visual direction

| Time | Visual |
|---|---|
| 0:00–0:03 | Big square of static clears into the cat astronaut; NOISE meter drops 100% → 0%. |
| 0:03–0:13 | RANDOM NOISE square appears on the axis, slides left as the arrow and "A CAT IN A SPACE SUIT" square arrive; DIFFUSION pill; "used by many image generators". |
| 0:13–0:23 | The clean picture gets noisier on each cue (0% → 25% → 60% → 100%) with a meter; a four-step strip fills in left to right (dashed placeholders keep it centered). |
| 0:23–0:36 | NOISY PICTURE → MODEL chip → "GUESS: THIS NOISE" square; then a 6×2 grid of many training images pops in ("A HUGE COLLECTION OF IMAGES"). |
| 0:36–0:52 | Fresh static; "STEP n / 40" counter; the image resolves coarse-to-fine (blurry blobs, then edges, then details) with SHAPES / EDGES / DETAILS pills on cue. |
| 0:52–1:03 | Prompt box types "a cat in a space suit" → TEXT ENCODER → a row of numbers → the image denoises below, with CAT and SPACE SUIT tags popping in on the words. |
| 1:03–1:17 | Three different pictures labeled SEED 7 / SEED 81 / SEED 2; then a row of three identical pictures, each SEED 42, and "SAME SEED = SAME IMAGE". |
| 1:17–1:27 | A folder icon gets a red × (vector) and fades; a final square is carved out of static into the image; visual end card. |

## On-screen text

IT STARTED AS PURE STATIC → HOW NOISE BECOMES A PICTURE → STEP 1: ADD NOISE UNTIL IT'S GONE → THE MODEL'S JOB: GUESS THE NOISE → STEP 2: REMOVE IT, A LITTLE AT A TIME → YOUR PROMPT STEERS EVERY STEP → SAME PROMPT, DIFFERENT STATIC → AN AI IMAGE IS CARVED FROM NOISE

## Ending

A clear takeaway line ("carved out of noise, one tiny cleanup at a time"). No like/follow ask in the voiceover; the end card is visual only.

## Thumbnail

"THIS STARTED / AS STATIC" (white/yellow) over one square image that is colorful static on the left and fades into the clean cat-astronaut picture on the right. Self-check passed (lines at 540.0, visual box 188–892).

## Captions & hashtags (all 3 platforms)

11:30 AM slot (Yangon) · Sun 11 Oct 2026 · A Bit Inside

Video: /workspace/abitinside/2026-10-11/video/02-ai-images-from-noise-final.mp4 (1:27.0, 1080x1920, H.264 + AAC)
Cover: /workspace/abitinside/2026-10-11/thumbnails/02-ai-images-from-noise-thumb.png (TikTok/Facebook) · 02-ai-images-from-noise-thumb.jpg (YouTube)

### TikTok (@abitinside)

Every AI picture you've seen started as random static 📺✨ Here's how diffusion actually works: the AI learns to spot noise, then removes it step by step until a picture appears. Plus why the same prompt never gives the same image twice.

#ai #aiart #howaiworks #techexplained #learnontiktok #ABitInside

### YouTube Shorts

Title: This AI Image Started as Pure Static: How AI Image Generators Work

Description:
Many popular AI image generators use diffusion. In training, the model sees pictures buried under more and more random noise and learns to guess which noise was added. To make a new image, it starts from fresh static and removes a little predicted noise at a time, often dozens of steps, so big shapes appear first and fine details last. Your prompt, turned into numbers by a text encoder, steers every step. Each run starts from different static, which is why the same prompt gives different pictures; reuse the same seed and settings and you can usually get the same image back. (The cat-astronaut picture and noise are illustrative animations, not output from a real AI model.)

Sources: Ho et al., "Denoising Diffusion Probabilistic Models" (2020) · Rombach et al., "Latent Diffusion Models" (2022) · Hugging Face Diffusers docs, "Reproducibility"

#Shorts #AI #AIArt #ABitInside

**Facebook Reels (A Bit Inside Page)**

Every AI-generated picture begins as random TV static. So how does it become a cat in a space suit? Here's how diffusion works, step by step, and why the same prompt gives you a different image every time.

#ABitInside #AI #TechExplained

## Sources


- Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models," NeurIPS 2020 (forward process gradually adds Gaussian noise "until signal is destroyed"; network trained to predict the added noise ε; sampling starts from pure Gaussian noise and removes predicted noise step by step, T = 1000 in their experiments; "large scale image features appear first and details appear last"): https://arxiv.org/abs/2006.11239
- Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models" (Stable Diffusion's method; denoising runs in a compressed latent space and is conditioned on text via cross-attention): https://arxiv.org/abs/2112.10752
- Lee et al., "Diffusion Explainer: Visual Explanation for Text-to-image Stable Diffusion" (2023) (prompt is tokenized and encoded into vectors by the CLIP text encoder; those representations guide the step-by-step denoising; guidance scale controls prompt adherence): https://arxiv.org/abs/2305.03509
- Jay Alammar, "The Illustrated Stable Diffusion" (step-by-step denoising of random latents; noise predictor; result decoded to an image): http://jalammar.github.io/illustrated-stable-diffusion/
- Hugging Face Diffusers docs, "Reproducibility" ("Diffusion is a random process that generates a different output every time"; pipelines use a different random seed each time to create the initial noisy tensors; setting a seed with a Generator reproduces results, within a tolerance and not guaranteed across all hardware): https://huggingface.co/docs/diffusers/using-diffusers/reusing_seeds
- Carlini et al., "Extracting Training Data from Diffusion Models" (2023), for the accuracy note on rare memorization: https://arxiv.org/abs/2301.13188


**Accuracy notes:** "Many popular image generators" is deliberate: not every AI image model is a diffusion model (some are autoregressive). Popular diffusion systems like Stable Diffusion denoise a compressed "latent" version of the image and decode it at the end; the video shows pixel static for clarity, an illustrative simplification. "Often dozens of times": DDPM used 1,000 steps, and common samplers today use roughly 20–50; the "STEP n / 40" counter is illustrative. "You can usually get the same image back" is hedged because Hugging Face notes reproducibility isn't guaranteed across hardware/software. "Isn't fetched from a folder of pictures" describes how generation works (no lookup or retrieval); research has shown models can occasionally reproduce near-copies of training images (Carlini et al. 2023), so we avoid saying "never copies". The cat-astronaut picture, the noise, the number row and the seed values are our own illustrative mock-ups (drawn as vectors and random noise in code), not output from any real AI model. No third-party logos.

## Scores (1–100)

| Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|
| 89 | 88 | 83 | 84 | 81 | 88 | 72 | **85** |

# Slot 03 · 2:30 PM (Yangon) · Curiosity "wait… WHAT?"

## Most Web Traffic Isn't Human: Bots Now Outnumber People Online




Cover thumbnail
`thumbnails/03-bots-outnumber-humans-thumb.jpg`

| Field | Detail |
|---|---|
| **Topic** | Bots now generate more web traffic than humans (Imperva's 2026 Bad Bot Report, 2025 data): what bots are, the good and the bad, and why CAPTCHAs exist |
| **Category** | Curiosity "wait… WHAT?" · Progressive Reveal (shocking claim → step-by-step truth) |
| **Format** | 10×10 "people vs bots" grid that flips, report card + split bar, browser mock-up, 100-cell traffic waffle, year bars, CAPTCHA mock-up |
| **Runtime** | 1:25.3 (85.3 s) · ffprobe-verified · within the 80–110 s brief and the 60–180 s rule |
| **Video** | `video/03-bots-outnumber-humans-final.mp4` (1080×1920, 30 fps, H.264 + AAC, −14 LUFS) |
| **Thumbnail** | `thumbnails/03-bots-outnumber-humans-thumb.png` (1080×1920) · `thumbnails/03-bots-outnumber-humans-thumb.jpg` (YouTube, 277 KB) |

## Why this topic

"Most of the web isn't human" sounds like the "dead internet" conspiracy theory, which is exactly why it works as a "wait… WHAT?": the reveal is that it's measured, and the numbers are fresh (Imperva's 2026 report, published April 29, 2026, covering 2025). It reveals step by step (what a bot is → the helpful 13% → the harmful 40% → it's growing → that's why you get CAPTCHAs) and lands on a personal twist ("you're in the minority"). It's new for the channel: Oct 9's AI-agents video was about how agents think, not about who makes up web traffic, and it doesn't overlap today's 01 (wireless EV road), 02 (diffusion), 04 (Year 2038) or 05 (Voyager 1).

## Hook (0–2 s)

- **Spoken:** "Most of the traffic on the web isn't human."
- **On-screen:** MOST WEB TRAFFIC ISN'T HUMAN
- **Visual:** a 10×10 grid of 100 people; on "isn't", 53 of them flip into robot heads.

## Full voiceover script

**[0:00 Hook]** Most of the traffic on the web isn't human.

**[0:02 The reveal]** And that's not the dead internet conspiracy theory. It's measured. Imperva's 2026 Bad Bot Report found that in 2025, bots made up fifty-three percent of the web traffic it observed. Humans, just forty-seven.

**[0:16 What is a bot]** A bot is just a program that uses websites by itself. No eyes, no hands. It loads pages, reads them, clicks and types, automatically, all day and all night.

**[0:28 The good ones]** Some bots are helpful. Search engines send crawlers to read web pages, so you can find them later. AI companies send crawlers too, to gather data for their models. Those good bots were thirteen percent of traffic.

**[0:42 The bad ones]** But forty percent came from bad bots. Programs that copy prices and content, try stolen passwords on one account after another, or grab concert tickets before real fans can.

**[0:54 It's growing]** And the bots are gaining ground. A year earlier, they were fifty-one percent. The report also says activity from AI-driven bots grew more than tenfold in a single year.

**[1:05 Why CAPTCHAs exist]** That's why websites keep asking you to prove you're human. The blurry letters, the photos of traffic lights, the I'm not a robot box. They're there to sort the people from the programs.

**[1:16 Ending]** So the next time you tick I'm not a robot, remember: on today's web, you're in the minority.

## Visual direction

| Time | Visual |
|---|---|
| 0:00–0:02 | 10×10 grid of orange people; 53 flip (card-turn) into cyan robot heads on "isn't". |
| 0:02–0:16 | "DEAD INTERNET THEORY" pill gets a red strike-through; green "MEASURED" pill; IMPERVA · 2026 BAD BOT REPORT card; split bar fills cyan 53% (bot icon) and orange 47% (person icon); "53% BOTS / 47% HUMANS / of web traffic Imperva observed, 2025". |
| 0:16–0:28 | A robot icon "drives" a browser mock-up: the page loads (progress bar), a scan line reads it, the cursor clicks GO (ripple), a password field fills with dots; "24 HOURS A DAY" pill. |
| 0:28–0:42 | Three page cards; a green crawler bot glides across, ticking each (vector checks); pills SEARCH ENGINE CRAWLERS and AI CRAWLERS; a 100-cell waffle fills 13 green cells. |
| 0:42–0:54 | Red bot; three dashed slots fill with red cards on cue: PRICES (price tag), PASSWORDS (lock + dots), TICKETS (ticket); waffle shows 40 red + 13 green of 100. |
| 0:54–1:05 | Two bars, BOTS' SHARE OF WEB TRAFFIC: 2024 51%, 2025 53%, over a dashed 50% line; pill "AI-DRIVEN BOTS: OVER 10x IN 2025". |

| Time | Visual |
|---|---|
| 1:05–1:16 | CAPTCHA mock-ups appear on cue: wavy letters, a 3×3 photo grid where traffic-light tiles get selected, and an "I'm not a robot" box that gets ticked; a green person and a red bot flank it. |
| 1:16–1:25 | The box is ticked again; the 100-icon grid returns, bots dim, and one person in the middle is ringed in yellow: "YOU: 1 OF THE 47%"; visual end card. |

## On-screen text

MOST WEB TRAFFIC ISN'T HUMAN → BOTS: 53%. HUMANS: 47%. → SO WHAT IS A BOT? → GOOD BOTS: 13% OF TRAFFIC → BAD BOTS: 40% OF TRAFFIC → AND THE BOTS ARE GAINING → WHY SITES ASK IF YOU'RE HUMAN → ON THE WEB, YOU'RE THE MINORITY

## Ending

A personal twist that resolves the hook ("next time you tick I'm not a robot… you're in the minority"). No like/follow ask in the voiceover; the end card is visual only.

## Thumbnail

"MOST WEB TRAFFIC / ISN'T HUMAN" (white, red, orange) over the 10×10 grid of cyan robot heads and orange people, with "53% BOTS. 47% HUMANS." in cyan below. Self-check passed (lines at 540.0, visual box 182–898).

## Captions & hashtags (all 3 platforms)

2:30 PM slot (Yangon) · Sun 11 Oct 2026 · A Bit Inside

Video: /workspace/abitinside/2026-10-11/video/03-bots-outnumber-humans-final.mp4 (1:25.3, 1080x1920, H.264 + AAC)
Cover: /workspace/abitinside/2026-10-11/thumbnails/03-bots-outnumber-humans-thumb.png (TikTok/Facebook) · 03-bots-outnumber-humans-thumb.jpg (YouTube)

### TikTok (@abitinside)

Most web traffic isn't human anymore 🤖 Imperva's 2026 Bad Bot Report found bots made up 53% of the traffic it saw in 2025: 13% helpful crawlers, 40% bad bots. That's why you keep ticking "I'm not a robot." Did you know you're the minority online?

#bots #internet #cybersecurity #techexplained #learnontiktok #ABitInside

### YouTube Shorts

Title: Most Web Traffic Isn't Human: Bots Now Outnumber People Online

Description:
In 2025, bots made up 53% of the web traffic Imperva observed, and humans just 47% (Imperva / Thales, 2026 Bad Bot Report). Good bots like search-engine and AI crawlers were 13%; bad bots that scrape prices and content, try stolen passwords, or scalp tickets were 40%. A year earlier, bots were 51%, and the report says AI-driven bot activity grew more than tenfold in 2025. That's why websites keep asking you to prove you're human. (Figures are for traffic Imperva observes, not the entire internet. Graphics are illustrative.)

Sources: Imperva (Thales), "2026 Bad Bot Report: Bad Bots in the Agentic Age" · Imperva blog, "Bad Bot Report 2026" (Apr 29, 2026)

#Shorts #Bots #Internet #ABitInside

**Facebook Reels (A Bit Inside Page)**

More than half of web traffic now comes from bots, not people. Here's what those bots actually are, which ones help you, which ones try to break into accounts or grab tickets, and why websites keep asking you to prove you're human. Next time you tick "I'm not a robot", remember: you're the minority online.

#ABitInside #Internet #TechExplained

## Sources

- Imperva (a Thales company), "2026 Thales Bad Bot Report: Bad Bots in the Agentic Age" (© Thales, April 2026). Executive summary and "Bad Bot vs Good Bot vs Human Traffic in 2025": "Automated traffic accounted for 53 percent of all observed web traffic in 2025. Of that, 40 percent was generated by bad bots, up from 37 percent the year before. The other 13 percent was benign automation, which includes harmless traffic such as search engines and AI crawlers. Human traffic, accounting for 47 percent..."; "Over the past 12 months, AI-driven bot activity has increased more than tenfold, with daily blocked requests rising from 2 million to 25 million in 2025"; bad bots include "scrapers, credential stuffing tools, and scalpers"; "Current analysis is based on detectable traffic". PDF: https://www.imperva.com/resources/reports/2026-Thales-Bad-Bot-Report.pdf
- Imperva blog, Tim Chang, "Bad Bot Report 2026: The Internet Is No Longer Human..." (Apr 29, 2026): "more than 53% of all web traffic in 2025, up from 51% the year before. Human activity has declined to just 47%": https://www.imperva.com/blog/bad-bot-report-2026-bots-agentic-age/

**Accuracy notes:** All percentages are Imperva's measurements of the web traffic it observes across its own network and customers, not a census of the entire internet; the voiceover says "of the web traffic it observed", and the on-screen line says "of web traffic Imperva observed, 2025". "Traffic" means requests to websites and apps, not "visitors" or "people". "Grew more than tenfold" is the report's own wording for AI-driven bot activity (daily blocked requests, 2M → 25M). The "dead internet theory" is mentioned only to say the finding is not that theory. The concert-ticket example illustrates scalping, which the report lists as a bad-bot activity. The 51% figure for 2024 is from Imperva's 2026 blog ("up from 51% the year before"). The CAPTCHA, browser, crawler and icon graphics are our own illustrative mock-ups, not any real company's product; no third-party logos (Imperva appears as plain text only).

## Scores (1–100)

| Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|
| 91 | 90 | 84 | 88 | 87 | 84 | 84 | **87** |

# Slot 04 · 5:30 PM (Yangon) · FLAGSHIP

## In 2038, Some Computers Will Think It's 1901: The Year 2038 Problem




Cover thumbnail
`thumbnails/04-year-2038-problem-thumb.jpg`

| Field | Detail |
|---|---|
| **Topic** | The Year 2038 problem: computers that count seconds since 1970 in a signed 32-bit number run out of room at 03:14:07 UTC on Jan 19, 2038, and wrap to Dec 13, 1901 |
| **Category** | Flagship · Systems explainer with a story (cause → countdown → overflow → real past failure → who's at risk → the fix) · Evergreen, with a live countdown and 2020–2025 fixes |
| **Format** | Glitching system clock, live seconds counter on a 1970→2038 timeline, 32-bit grid that fills and flips, date cards, AOLserver timeline story, device-risk cards, 32→64-bit expansion |
| **Runtime** | 2:26.6 (146.6 s) · ffprobe-verified · within the 110–150 s flagship brief and the 60–180 s rule |
| **Video** | `video/04-year-2038-problem-final.mp4` (1080×1920, 30 fps, H.264 + AAC, −14 LUFS) |
| **Thumbnail** | `thumbnails/04-year-2038-problem-thumb.png` (1080×1920) · `thumbnails/04-year-2038-problem-thumb.jpg` (YouTube, 149 KB) |

## Why this topic

It's a real, dated deadline that almost nobody outside software has heard of, and it explains something genuinely "inside" every device: computers don't store dates, they count seconds. It has everything a flagship needs: a shocking hook (2038 → 1901), a clear mechanism you can see (32 boxes filling up, then flipping), a countdown that makes it personal (we've already used ~83% of the count), a true story where it already broke software (AOLserver, May 13, 2006), a fair answer to "should I worry?", and a satisfying fix with a mind-bending number (292 billion years). The most research of the day: every date and number was checked against primary or well-sourced references, and the derived ones were computed. Fresh for the channel (no time-overflow

topic on Oct 8–10; Oct 9's GPS video was about relativity) and no overlap with today's 01 (EV road), 02 (diffusion), 03 (bots) or 05 (Voyager 1).

## Hook (0–2 s)

- **Spoken:** "In 2038, some computers will think it's 1901."
- **On-screen:** IN 2038, SOME COMPUTERS WILL THINK IT'S 1901
- **Visual:** a SYSTEM CLOCK card reading JAN 19, 2038 · 03:14:07 UTC glitches (RGB split) and becomes DEC 13, 1901 · 20:45:52 UTC in red on "1901".

## Curiosity gap (3–11 s)

"Not because of a virus. Not because of a hacker. Just one number, running out of room." Two struck-through pills (A VIRUS, A HACKER), then a 32-bit grid fills up and a "ROOM USED" bar goes red: FULL.

## Full voiceover script

**[0:00 Hook]** In 2038, some computers will think it's 1901.

**[0:03 Setup]** Not because of a virus. Not because of a hacker. Just one number, running out of room. It's called the Year 2038 problem.

**[0:11 How computers count time]** Many computers don't store dates the way we do. They count seconds: how many have passed since midnight on January 1st, 1970, a moment programmers call the Unix epoch. Right now, that count is about 1.79 billion.

**[0:27 The 32-bit limit]** For decades, lots of systems kept that count in a signed 32-bit number. Thirty-two binary digits, with one of them used for the plus or minus sign. So the biggest value it can hold is 2,147,483,647.

**[0:46 The last second]** Count that many seconds from 1970, and you land on January 19th, 2038, at 3:14 and 7 seconds, UTC. We've already used about eighty-three percent of the count.

**[1:00 The flip]** One second later, the number doesn't grow. It flips to the most negative value it can hold. And a computer reading it sees: December 13th, 1901.

**[1:11 It already happened]** It has already bitten once. In 2006, a web server program called AOLserver gave some requests a timeout of one billion seconds, its way of saying never. But on May 13th, 2006, now plus one billion seconds reached past 2038. The math overflowed, and the software started crashing.

**[1:32 Who's at risk]** Most modern phones and computers already count time with 64 bits. The real risk is in machines that rarely get updated: embedded computers in vehicles, factories, routers and cameras, built to run for decades.

**[1:47 The fix]** The fix is a bigger number. In 2020, Linux 5.6 made the kernel ready for 32-bit systems to run past 2038. And in 2025, Debian moved nearly all of its systems to 64-bit time. A 64-bit count won't run out for about 292 billion years.

**[2:07 Ending + CTA]** So the fix is known, and the deadline is fixed. What's left is the quiet part: finding every old machine still counting in 32 bits, before its last second ticks by. Follow A Bit Inside to see what's really happening inside the tech you use every day.

## Visual direction

| Time | Visual |
|---|---|
| 0:00–0:03 | SYSTEM CLOCK date card: JAN 19, 2038 / 03:14:07 UTC glitches into red DEC 13, 1901 / 20:45:52 UTC. |
| 0:03–0:11 | Pills A VIRUS and A HACKER pop in and get red strike-throughs; a 4×8 grid of 32 bits fills with 1s (purple sign bit + 31 cyan bits); a ROOM USED bar fills to FULL (red); yellow pill THE YEAR 2038 PROBLEM. |
| 0:11–0:27 | Timeline 1970 → 2038; a counter races from 0 to the real current Unix time (1,791,676,8xx) and keeps ticking each second, a yellow NOW dot at ~83% of the line; pill JAN 1, 1970 = THE UNIX EPOCH; "about 1.79 billion". |
| 0:27–0:46 | Pill SIGNED 32-BIT NUMBER; 32 boxes, the first labelled +/- SIGN, "31 BITS FOR THE NUMBER"; on "biggest" every number bit flips to 1 in sequence; MAXIMUM VALUE card counts up to 2,147,483,647. |
| 0:46–1:00 | Progress bar 1970 → 2038 fills; date card THE LAST SECOND 32 BITS CAN COUNT: JAN 19, 2038 · 03:14:07 UTC; a TODAY marker at 83%; "~83% ALREADY USED · 1,791,676,800 of 2,147,483,647 seconds". |
| 1:00–1:11 | The 32-bit grid flips from 0111…1 to 1000…0 (sign bit turns red); the number glitches from 2,147,483,647 to −2,147,483,648 (MOST NEGATIVE VALUE); date card WHAT THE COMPUTER READS: DEC 13, 1901. |
| 1:11–1:32 | Server rack labelled AOLserver; code card "timeout = now + 1,000,000,000 s"; "MAY 13, 2006: NOW + 1 BILLION SECONDS"; timeline 1901 · 2006 · 2038 where an orange arc from 2006 lands just past 2038, then a red dashed overflow arc wraps back to 1901 ("OVERFLOW: DEADLINE IN THE PAST"); the server turns red and gets a vector ✕ on "crashing". |
| 1:32–1:47 | Phone and laptop with a green check: MOST MODERN DEVICES: 64-BIT; four dashed slots fill with yellow cards on cue: VEHICLES, FACTORIES, ROUTERS, CAMERAS; pill RARELY UPDATED, BUILT TO LAST. |
| 1:47–2:07 | The 32-bit grid doubles into 64 green bits; pills 2020: LINUX 5.6 READY FOR 2038+ and 2025: DEBIAN SWITCHES TO 64-BIT TIME; "292 BILLION YEARS · ~21x the age of the universe". |
| 2:07–2:26 | Card TIME LEFT ON THE 32-BIT CLOCK: ~11 YEARS (about 355 million seconds); five device cards flip from 32 (yellow) to 64 (green) one by one on "finding every old machine"; a green stopwatch sweeps; Follow/Subscribe end card. |

## On-screen text

IN 2038, SOME COMPUTERS WILL THINK IT'S 1901 → ONE NUMBER, OUT OF ROOM → COMPUTERS COUNT SECONDS SINCE 1970 → THE 32-BIT NUMBER LIMIT → THE LAST SECOND: JAN 19, 2038 → ONE TICK LATER: IT'S 1901 → IT ALREADY HAPPENED ONCE → WHO'S ACTUALLY AT RISK? → THE FIX: A BIGGER NUMBER → THE DEADLINE IS FIXED

## Call to action (slot 04 only)

A natural, earned CTA after the story resolves ("the fix is known… before its last second ticks by"): "Follow A Bit Inside to see what's really happening inside the tech you use every day." Spoken once at the end, plus the visual end card.

## Thumbnail

"IN 2038, / IT'S 1901" (white, cyan, red) over two stacked date cards: JAN 19, 2038 (cyan, "+1 SECOND"), a vector down-arrow, then a glitched red DEC 13, 1901 ("32-BIT OVERFLOW") on a red glow. Self-check passed (lines at 540.0, visual box 148–932).

## Captions & hashtags (all 3 platforms)

5:30 PM slot (Yangon) · Sun 11 Oct 2026 · A Bit Inside

Video: /workspace/abitinside/2026-10-11/video/04-year-2038-problem-final.mp4 (2:26.6, 1080x1920, H.264 + AAC)
Cover: /workspace/abitinside/2026-10-11/thumbnails/04-year-2038-problem-thumb.png (TikTok/Facebook) · 04-year-2038-problem-thumb.jpg (YouTube)

### TikTok (@abitinside)

On January 19, 2038 at 03:14:07 UTC, some computers will think it's 1901 ⏰ Many systems count time as seconds since 1970, and a 32-bit number tops out at 2,147,483,647. It already crashed software once, back in 2006. Here's who's at risk and how it gets fixed. Follow for more of what's inside your tech.

#y2038 #programming #computerscience #techexplained #learnontiktok #ABitInside

### YouTube Shorts

Title: In 2038, Some Computers Will Think It's 1901: The Year 2038 Problem

Description:
Many computers store time as the number of seconds since January 1, 1970 (Unix time). Systems that keep that count in a signed 32-bit number max out at 2,147,483,647, which is 03:14:07 UTC on January 19, 2038. One second later the number flips negative, and the date reads December 13, 1901. It has already bitten once: in May 2006, AOLserver's one-billion-second timeouts overflowed and crashed. Most modern 64-bit devices are fine; the risk is in rarely updated embedded systems. The fix is 64-bit time (Linux 5.6 in 2020, Debian in 2025), which lasts about 292 billion years. (Graphics are illustrative; date math is our own calculation.)

Sources: Wikipedia, "Year 2038 problem" · Linux kernel y2038 pull request (Arnd Bergmann, 2020) · Debian 13 release notes, "64-bit time_t ABI transition"

#Shorts #Programming #Technology #ABitInside

### Facebook Reels (A Bit Inside Page)

Your devices don't really store the date. Many of them just count seconds since January 1, 1970, and in some machines that counter runs out on January 19, 2038. One second later, they could think it's 1901. Here's how that happens, the time it already crashed real software, who's actually at risk, and how engineers are fixing it. Follow A Bit Inside for more of what's happening inside the tech you use every day.

#ABitInside #Technology #TechExplained

## Sources

- Wikipedia, "Year 2038 problem" (checked Oct 11, 2026; cites primary sources): Unix time = seconds since 00:00:00 UTC, Jan 1, 1970; signed 32-bit maximum 2^31 − 1 = 2,147,483,647; overflow to the minimum value, read as a date in the past; most vulnerable are rarely updated legacy and embedded systems (transport, routers, wireless access points, IP cameras, 32-bit file systems and databases); 64-bit time lasts ~292 billion years, ~21× the age of the universe; Linux 5.6 (2020) supports 64-bit time_t on 32-bit architectures; glibc 2.34 (Aug 2021) adds 64-bit time_t on 32-bit platforms; "Early problems": in May 2006, AOLserver's default maximum time-out of one billion seconds overflowed for requests after 01:27:28 UTC on 13 May 2006, returning dates in the past and crashing the software: https://en.wikipedia.org/wiki/Year_2038_problem
- Arnd Bergmann, "[GIT PULL] y2038: core, driver and file system changes" (Linux kernel mailing list, Jan 29, 2020; tag y2038-drivers-for-v5.6: "the first release that can serve as a base for a 32-bit system designed to run beyond year 2038, with a few remaining caveats: All user space must be compiled with a 64-bit time_t"): https://lkml.iu.edu/hypermail/linux/kernel/2001.3/05499.html · Phoronix, "Linux 5.6 Is The First Kernel For 32-Bit Systems Ready To Run Past Year 2038": https://www.phoronix.com/news/Linux-5.6-32-bit-Past-Y2038
- Debian 13 "trixie" release notes, §2.2.6 "64-bit time_t ABI transition" ("All architectures other than i386 now use a 64-bit time_t ABI, supporting dates beyond 2038"): https://www.debian.org/releases/trixie/release-notes/whats-new.en.html
- Original AOLserver report cited by Wikipedia: "The Future Lies Ahead" (June 28, 2006), archived: https://web.archive.org/web/20061128235428/http://substitute.livejournal.com/1430908.html

**Derived numbers (our arithmetic, computed in Python):** 1970-01-01 + 2,147,483,647 s = 2038-01-19 03:14:07 UTC; −2,147,483,648 s = 1901-12-13 20:45:52 UTC; Unix time at 2026-10-11 00:00 UTC = 1,791,676,800 → "about 1.79 billion" and 1,791,676,800 ÷ 2,147,483,647 = 83.4% → "about eighty-three percent"; 2,147,483,647 − 1,791,676,800 = 355,806,847 s ≈ 11.3 years → "~11 YEARS / about 355 million seconds"; 2^63 s ÷ 31,556,952 s per year ≈ 292.3 billion years; 2038-01-19 03:14:07 − 1,000,000,000 s = 2006-05-13 01:27:27 UTC (the first request whose timeout landed past the limit, matching the reported 01:27:28 cut-off); 10^9 s ≈ 31.7 years.

**Accuracy notes:** "Some computers" and "lots of systems" are deliberate: only systems that still store time as a signed 32-bit count are affected, and most modern 64-bit phones and computers already use 64-bit time (Windows tracks time differently and isn't affected by this specific bug). The Linux 5.6 line says the kernel became ready; 32-bit programs still need to be rebuilt with 64-bit time, which is why Debian's 2025 transition mattered. "Nearly all of its systems" = all Debian architectures except i386, which was left on 32-bit time for legacy software. The AOLserver story is the widely reported early case; the crash affected servers running that default setting until operators lowered the timeout. Vehicles, factories, routers and cameras are examples of long-lived embedded systems named in the references as potentially at risk, not specific products known to fail. The live counter in 0:11–0:27 starts from Unix time at 00:00 UTC on Oct 11, 2026 (illustrative "now"). All graphics are our own illustrative mock-ups; no third-party logos (Linux, Debian and AOLserver appear as plain text only).

## Scores (1–100)

| Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|
| 93 | 92 | 87 | 89 | 86 | 89 | 80 | **89** |

# Slot 05 · 8:30 PM (Yangon) · Deep / Reflective

## Voyager 1 Will Soon Be One Light-Day From Home




Cover thumbnail
`thumbnails/05-voyager-one-light-day-thumb.jpg`

| Field | Detail |
|---|---|
| **Topic** | On Nov 18, 2026, Voyager 1 becomes the first human-made object one light-day from Earth: what that distance means, the 1977 machine still answering, its fading power, the Pale Blue Dot and the Golden Record |
| **Category** | Deep / emotional / future-of-life · calm, reflective, closed ending (no cliffhanger) |
| **Format** | Starfield scenes on one axis: Earth-to-Voyager light line with a travelling light pulse, year counter 1977 → 2026, message/reply pulses, memory-vs-photo scale, power bar + 10-instrument status grid, Pale Blue Dot frame, spinning golden record, drifting spacecraft with a distant star |
| **Runtime** | 1:33.3 (93.3 s) · ffprobe-verified · within the 80–120 s brief and the 60–180 s rule |
| **Video** | `video/05-voyager-one-light-day-final.mp4` (1080×1920, 30 fps, H.264 + AAC, −14 LUFS) |
| **Thumbnail** | `thumbnails/05-voyager-one-light-day-thumb.png` (1080×1920) · `thumbnails/05-voyager-one-light-day-thumb.jpg` (YouTube, 121 KB) |

## Why this topic

It's tied to the news: NASA says Voyager 1 hits one light-day from Earth on Nov 18, 2026, so the video is timely now and will be shareable again on the day. It fits the evening slot well. The facts are concrete (a day for a message to arrive, about 68 KB of memory, 2 of 10 instruments still on), and the emotional pull comes from the Pale Blue Dot and the Golden Record. The ending is closed and reflective rather than a cliffhanger. It's new for the channel: nothing from Oct 8–10 covered space, and it doesn't overlap today's 01 (wireless EV road), 02 (diffusion), 03 (bots) or 04 (Year 2038).

## Hook (0–4 s)

- **Spoken:** "On November 18th, something we built will be one full light-day from home."
- **On-screen:** ONE LIGHT-DAY FROM HOME
- **Visual:** Voyager at the top and Earth at the bottom of a dotted line, with a light pulse climbing from Earth to the spacecraft. Pills: NOV 18, 2026 and 1 LIGHT-DAY AWAY.

## Full voiceover script

**[0:00 Hook]** On November 18th, something we built will be one full light-day from home.

**[0:05 Voyager 1]** It's Voyager 1. It left Earth in 1977, and it has been flying away from us ever since. Forty-nine years, and counting.

**[0:14 The distance]** A light-day is how far light travels in 24 hours: about 26 billion kilometers. Send Voyager a message, and it takes a whole day to arrive. Its answer takes another day to come back.

**[0:27 The machine]** And it still answers. Its computers hold about 68 kilobytes of memory, all together. Less than one ordinary photo on your phone.

**[0:36 Fading power]** Its power comes from decaying plutonium, and it loses about four watts every year. So, one by one, the team has switched things off. Of its ten science instruments, two are still on.

**[0:49 Pale Blue Dot]** Before its cameras went dark in 1990, it turned around one last time and photographed Earth from about 6 billion kilometers away. We were about one pixel. Carl Sagan called it a pale blue dot.

**[1:02 Golden Record]** It also carries a golden record: greetings in 55 languages, music, the sounds of life on Earth, and 115 pictures of who we are.

**[1:13 Ending]** Someday, perhaps in the 2030s, its radio will go quiet. But Voyager will keep going. In about 40,000 years, it will pass another star, still carrying our songs. A message from a small blue dot, sent into the dark, just in case someone is listening.

## Visual direction

| Time | Visual |
|---|---|
| 0:00–0:05 | Mirrored starfield. Voyager (vector dish, booms, antennas) at the top and a glowing Earth at the bottom, joined by a dotted line. A yellow light pulse climbs the line. Pills: NOV 18, 2026, then a filled 1 LIGHT-DAY AWAY. |
| 0:05–0:14 | Large Voyager drifting gently. A LAUNCHED card shows 1977, then the year counter rolls to 2026 (STILL FLYING). Pill: 49 YEARS AND COUNTING. |
| 0:14–0:27 | Voyager at the top and Earth at the bottom, joined by the light line. Pill: ~26 BILLION KM. A cyan pulse travels up (MESSAGE OUT: 24 HOURS), then a yellow pulse travels back down (REPLY BACK: 24 HOURS). |
| 0:27–0:36 | A small glowing cyan square stands for Voyager's whole memory (VOYAGER'S MEMORY: ~68 KB). A photo frame then expands around it (ONE PHONE PHOTO: A FEW MB, "rough scale"). |
| 0:36–0:49 | A power bar labelled POWER FROM DECAYING PLUTONIUM slowly drains, with "-4 W EVERY YEAR". A 5×2 grid of the 10 science instruments (ISS, IRIS, UVS, PPS, PRA, PLS, LECP, CRS, MAG, PWS) goes |

| Time | Visual |
|---|---|
| | dark one by one until only MAG and PWS stay green. Pill: 2 OF 10 STILL ON. |
| 0:49–1:02 | Voyager with CAMERAS: LAST PICTURES. Cut to a dark photo frame with diagonal sunlight streaks and a single pale-blue pixel, which a yellow ring closes in on. Labels: "FEB 14, 1990 ~6 BILLION KM" and the pill PALE BLUE DOT. |
| 1:02–1:13 | A slowly spinning gold disc with grooves and a highlight sweep. Pills stack on cue: GREETINGS IN 55 LANGUAGES, MUSIC, SOUNDS OF LIFE ON EARTH, 115 PICTURES. |
| 1:13–1:33 | Faint radio arcs above Voyager fade to nothing ("quiet"). The craft drifts upward through a denser starfield. A distant yellow star twinkles in, with the pill IN ~40,000 YEARS: ANOTHER STAR. A tiny Earth glows at the bottom, labelled HOME. Then the visual end card. |

## On-screen text

ONE LIGHT-DAY FROM HOME → VOYAGER 1, LAUNCHED IN 1977 → ONE MESSAGE, ONE WHOLE DAY → STILL ANSWERING, ON ABOUT 68 KB → SWITCHING OFF, ONE BY ONE → WE WERE ONE PIXEL → A GOLDEN RECORD OF WHO WE ARE → STILL SAILING, JUST IN CASE

## Ending

A calm, complete close: the radio will one day go quiet, but the spacecraft keeps carrying our record outward, "just in case someone is listening." There is no open question and no cliffhanger. There's no like/follow ask in the voiceover; the end card is visual only.

## Thumbnail

"ONE LIGHT-DAY / FROM HOME" in white, cyan and yellow. Below it, a large vector Voyager in a blue glow over a mirrored starfield, a dotted line down to a glowing Earth, and a yellow pill reading "VOYAGER 1 - NOV 18, 2026". Self-check passed: lines at 540.0, visual box x 176–904, y 649–1392 (above 1420).

## Captions & hashtags (all 3 platforms)

8:30 PM slot (Yangon) · Sun 11 Oct 2026 · A Bit Inside

Video: /workspace/abitinside/2026-10-11/video/05-voyager-one-light-day-final.mp4 (1:33.3, 1080x1920, H.264 + AAC)
Cover: /workspace/abitinside/2026-10-11/thumbnails/05-voyager-one-light-day-thumb.png (TikTok/Facebook) · 05-voyager-one-light-day-thumb.jpg (YouTube)

### TikTok (@abitinside)

On November 18, something humans built will be a full light-day from Earth 🛰️ Voyager 1 left in 1977. Now a message takes a whole day to reach it, it still answers with about 68 KB of memory, and it carries a golden record of who we are. What would you put on that record?

#voyager1 #nasa #space #spaceexploration #learnontiktok #ABitInside

### YouTube Shorts

Title: Voyager 1 Will Soon Be One Light-Day From Home

Description:
On November 18, 2026 (2:16:07 a.m. PST), NASA's Voyager 1 will be one light-day from Earth: about 16.1 billion miles (25.9 billion km), the distance light travels in 24 hours, and the first human-made object to get that far. It launched in 1977. A message now takes a day to reach it, and its reply takes another day to come back. Its computers have about 68 KB of memory in total, its plutonium power source gives about 4 watts less every year, and only two of its ten science instruments are still on. In 1990 it took the Pale Blue Dot, where Earth is about a pixel in size, and it carries the Golden Record: greetings in 55 languages, music, natural sounds and 115 images. In about 40,000 years it will pass within about 1.7 light-years of the star Gliese 445. (Graphics are illustrative.)

Sources: NASA, "Voyager 1: What Is a Light-Day?" · NASA, "Where Are Voyager 1 and 2 Now?" · NASA Voyager FAQ · NASA, "Voyager 1's Pale Blue Dot" · NASA, "Golden Record Contents"

#Shorts #Voyager1 #Space #ABitInside

**Facebook Reels (A Bit Inside Page)**

On November 18, a spacecraft launched in 1977 will be one full light-day from Earth. Voyager 1 still answers our messages with about 68 KB of memory, on fading power, with just two of its ten science instruments still on. Here's the quiet story of how far it has gone, the Pale Blue Dot it sent home, and the golden record it carries into the dark.

#ABitInside #Voyager1 #Space

## Sources

- NASA Science, "Voyager 1: What Is a Light-Day?":
- "one light-day away from Earth. That's the distance a particle of light can travel in 24 hours, equaling about 16 billion miles (about 26 billion kilometers), a point that Voyager 1 will reach on Wednesday, Nov. 18, 2026, at 2:16:07 a.m. PST"
- "16,094,799,096 miles (25,902,068,356 kilometers)"
- "will have taken 49 years, two months, and 11 days"
- "cruising at nearly 38,000 mph (61,000 kph)"
- Voyager 1 "is expected to pass within about 1.7 light-years of [AC +79 3888 / Gliese 445] — in 40,000 years"
- https://science.nasa.gov/mission/voyager/voyager-1/voyager-1-what-is-a-light-day/
- NASA Science, "Where Are Voyager 1 and 2 Now?":
- Nov 18, 2026 distance of 16.094 billion miles (25.902 billion km), "the first human-made object to reach this distance from Earth"
- Instrument status table: ISS cameras off Feb 14, 1990; LECP off April 17, 2026; IRIS, PRA, UVS, CRS off to save power; PPS and PLS off because of degraded performance
- https://science.nasa.gov/mission/voyager/where-are-voyager-1-and-voyager-2-now/
- NASA Science, Voyager 1 mission page: "On April 17 … shut down an instrument aboard Voyager 1, to save power … Voyager 1 still has two science instruments operating" (MAG and PWS). https://science.nasa.gov/mission/voyager/voyager-1/
- NASA Science, Voyager FAQ: https://science.nasa.gov/mission/voyager/frequently-asked-questions/
- Memory: "a total of about 68KB"
- Power: the RTG "puts out 4 watts less each year … Heaters and other systems have also been turned off one by one"

- Lifetime: the spacecraft "could remain in the range of the Deep Space Network through about 2036, depending on how much power the spacecraft still have"
- NASA Science, "Voyager 1's Pale Blue Dot": https://science.nasa.gov/resource/voyager-1s-pale-blue-dot/
- Taken "Feb. 14, 1990 ... at a distance of 3.7 billion miles (6 billion kilometers) from the Sun"
- "Earth was just a point of light about a pixel in size"
- Taken "just 34 minutes before Voyager 1 powered off its cameras forever"
- Inspired Carl Sagan's book "Pale Blue Dot"
- NASA Science, "Golden Record Contents": https://science.nasa.gov/mission/voyager/golden-record-contents/
- "natural sounds, such as those made by surf, wind and thunder, birds, whales"
- "musical selections from different cultures and eras"
- "spoken greetings from Earth-people in fifty-five languages"
- "The 115 images are encoded in analog form"

**Accuracy notes:** - "Forty-nine years" is the time since its September 1977 launch; NASA puts it at 49 years, 2 months and 11 days on Nov 18. - "About 26 billion kilometers" is NASA's rounding of 25.9 billion km. - The two-day round trip for a message and reply is our own arithmetic from the one-light-day distance. - "Less than one ordinary photo on your phone" is our comparison: 68 KB against a typical phone photo of a few megabytes, and the on-screen scale is labelled "rough". - "Of its ten science instruments, two are still on" follows NASA's status table. Not every instrument was turned off for power: PPS and PLS failed. The script says only "switched things off", which is true. - "From about 6 billion kilometers away": NASA gives that distance from the Sun. At that range Earth's distance is essentially the same, and the script says "about". - "Perhaps in the 2030s, its radio will go quiet" is hedged on NASA's "through about 2036, depending on how much power". - "Pass another star" in about 40,000 years means within about 1.7 light-years of Gliese 445, per NASA. The YouTube description spells this out. - All graphics are our own vector illustrations. There are no NASA logos and no third-party branding.

## Scores (1–100)

| Hook | Curiosity | Retention | Shareability | Comment | Visual | Freshness | Overall viral |
|---|---|---|---|---|---|---|---|
| 87 | 85 | 84 | 88 | 82 | 85 | 89 | **86** |

### 🏆 Post of the Day: 04 · "In 2038, Some Computers Will Think It's 1901" (5:30 PM flagship)

It has the strongest hook of the day: a concrete, absurd-sounding date flip that turns out to be simple arithmetic. It's also the deepest story. It covers why computers count seconds since 1970, the 2,147,483,647 ceiling, the exact second it runs out, a real crash it already caused (AOLserver, 2006), who is actually at risk, and the 64-bit fix (Linux 5.6 in 2020, Debian in 2025). Every number is either sourced or our own clearly-labelled calculation, and it ends on a natural call to follow. **Runner-up:** 03, "Most web traffic isn't human". It's the best pure "wait… WHAT?" reveal, backed by brand-new 2026 data and a personal twist ("you're the minority online") that drives comments.